Homework 3

Artificial Intelligence in Robotics

ENGR 6010 - 01

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Program Explanation:

This program is a supervised classifier benchmark that trains five classic machine learning models on the Iris dataset and compares how well each one identifies flower species it has never seen before. The five models are Logistic Regression, K-Nearest Neighbors (K = 5), a Support Vector Machine with a Gaussian RBF kernel, a Random Forest with 100 trees, and Gaussian Naive Bayes. The core question the script answers is whether different classification methods, each built on different assumptions about the data, actually perform differently on the same problem when tested under identical conditions.

The process begins by loading the Iris dataset, which contains 150 flowers split evenly across three species (setosa, versicolor, and virginica), each described by four measurements: sepal length, sepal width, petal length, and petal width. The data is then split with train_test_split() into 105 training flowers and 45 testing flowers. I kept random_state=42 so the split is identical every time the script runs, which makes the results reproducible and guarantees every model is trained and tested on exactly the same flowers.

Each model takes a different approach to drawing boundaries between species. Logistic Regression learns linear decision boundaries, KNN labels a flower by majority vote among its 5 closest training samples, the SVM uses the RBF kernel to fit curved, nonlinear boundaries, Random Forest combines the votes of 100 decision trees, and Gaussian Naive Bayes calculates the probability of each species assuming every feature is independent within a class. A loop runs through all five models, calling fit() on X_train and y_train so each model learns the relationship between measurements and species, then calling predict() on X_test to classify the 45 unseen flowers.

The predictions are compared against the true labels in y_test using four metrics. Accuracy is the fraction of all test flowers labeled correctly. Precision and recall are calculated per species, where:

Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1 = 2 × (Precision × Recall) / (Precision + Recall)

Since Iris has three classes instead of two, I used average="macro", which computes each metric separately for every species and averages them so each species counts equally. The script also prints a classification_report() for each model to show the per-species breakdown. Finally, the results are collected into a pandas DataFrame, sorted by F1-score, and printed to four decimal places as the benchmark table below.

```
======================================================================
Classifier Benchmark on Iris (test set, n = 45, macro-averaged)
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                     Accuracy  Precision  Recall  F1-Score
Model
Logistic Regression    1.0000     1.0000  1.0000    1.0000
KNN (K=5)              1.0000     1.0000  1.0000    1.0000
SVM (RBF)              1.0000     1.0000  1.0000    1.0000
Random Forest          1.0000     1.0000  1.0000    1.0000
Gaussian Naive Bayes   0.9778     0.9762  0.9744    0.9743
======================================================================
```

Results:

Four of the five models classified all 45 test flowers correctly, scoring 1.0000 on every metric, while Gaussian Naive Bayes got 44 out of 45 for an accuracy of 0.9778 and a macro F1-score of 0.9743. The reason so many models hit a perfect score is that Iris is a fairly separable dataset. Setosa can be split from the other two species using petal measurements alone, and versicolor and virginica only overlap slightly, so on this split linear, distance-based, kernel-based, and tree-based methods all found boundaries that worked.

The classification_report() shows exactly where Naive Bayes went wrong: one versicolor flower was predicted as virginica. That single mistake dropped versicolor's recall to 0.9231 and virginica's precision to 0.9286. The error comes from the model's independence assumption. Petal length and petal width are strongly correlated in Iris, but Naive Bayes treats them as separate pieces of evidence, which skews its probability estimates right where versicolor and virginica overlap.

These results also have limits. With only 45 test flowers, one wrong prediction moves accuracy by about 2.22 percentage points, and the four-way tie only reflects this single split, not proof that the models are equally good. Running 10-fold cross-validation and reporting the mean and standard deviation would give a more reliable ranking. Adding StandardScaler in a pipeline would also matter for Logistic Regression, KNN, and SVM on other datasets, since those methods are sensitive to feature scale. It didn't change anything here because all four Iris measurements are in centimeters with similar ranges.